

Stable sparse features recover recognizable intents

Three validation-selected features evaluated on held-out Arabic and Chinese.

TAKEAWAY ORDER

feature 299

purity@20 0.60

stability 0.69

zh-CN

a=15.4

“order pizza for delivery”

ar-SA

a=13.8

“does pizza hut have delivery”

zh-CN

a=12.9

“hey order two wings with french fries from...”

IOT COFFEE

feature 1423

purity@20 1.00

stability 0.91

ar-SA

a=13.1

“olly turn on the coffee machine to make me...”

zh-CN

a=12.1

“make a coffee”

ar-SA

a=10.9

“coffee”

QA CURRENCY

feature 650

purity@20 0.95

stability 0.89

zh-CN

a=19.3

“dollar to euro exchange rate”

zh-CN

a=18.6

“what is the exchange rate between us and m...”

zh-CN

a=18.2

“what is the exchange rate of us dollar to...”